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(54) **DC-TO-DC CONVERTER AND METHOD FOR CONTROLLING SAME**

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See application file for complete search history.

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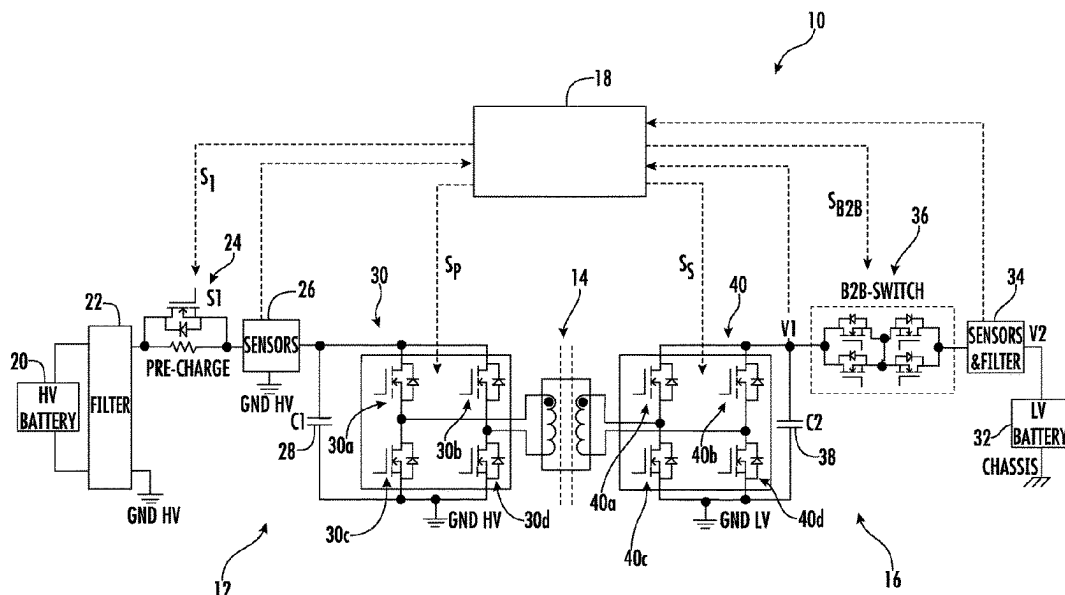
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(57) **ABSTRACT**

The present disclosure provides a converter including a primary stage having a plurality of primary stage switches, a secondary stage having a plurality of secondary stage switches and a capacitor, and a controller that isolates the secondary stage from a second battery, and that controls switching of the plurality of primary stage switches and the plurality of secondary stage switches to provide a plurality of bursts of energy from a first battery to the capacitor to pre-charge the capacitor to a threshold voltage, wherein after an initial burst of the plurality of bursts having an initial duration, a subsequent burst of the plurality of bursts provided has a duration equal to a duration of a preceding burst plus a time period based on a voltage of the first battery.

20 Claims, 3 Drawing Sheets



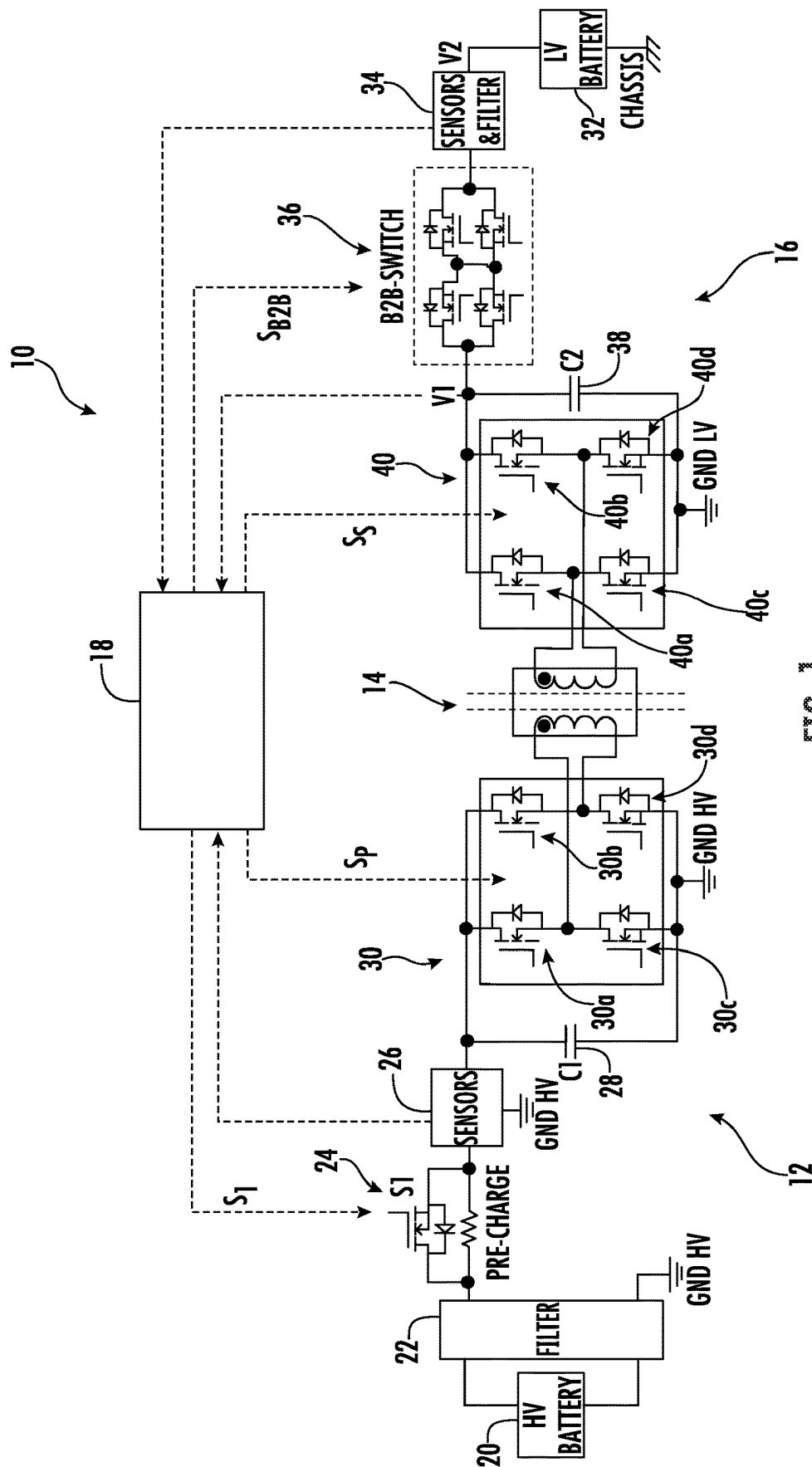
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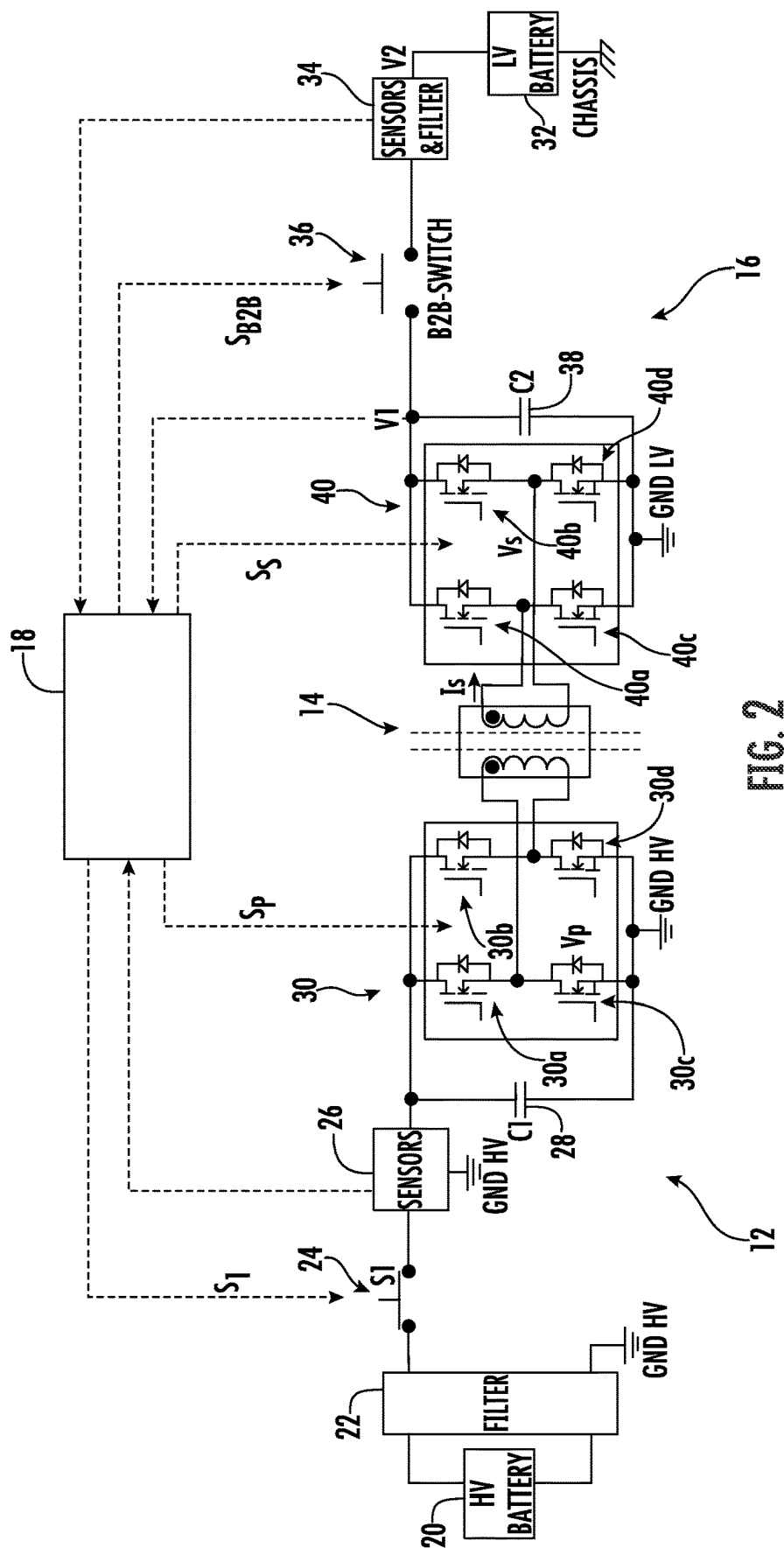


FIG. 2

HV RANGE [Vdc]	RATE t1
430 TO 470	50ns/ms
290 TO 430	470ns/ms TO 50ns/ms
250 TO 300	100ns/ms

FIG. 3

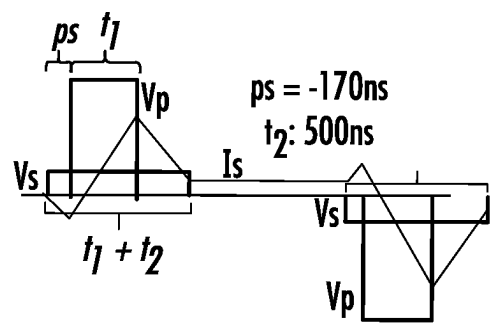


FIG. 4

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DC-TO-DC CONVERTER AND METHOD FOR CONTROLLING SAME

TECHNICAL FIELD

The present disclosure relates to a DC-to-DC converter and a method for controlling a DC-to-DC converter.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a simplified electrical schematic/block diagram of a non-limiting, exemplary embodiment of a DC-to-DC converter;

FIG. 2 is a simplified electrical schematic/block diagram of a non-limiting, exemplary embodiment of a DC-to-DC converter;

FIG. 3 is a table of selected time periods corresponding to battery voltage ranges in a non-limiting, exemplary embodiment according to the present application; and

FIG. 4 is a simplified timing and signal diagram illustrating exemplary charging pulses in a non-limiting, exemplary embodiment according to the present disclosure.

DETAILED DESCRIPTION

Reference will now be made in detail to embodiments, examples of which are illustrated in the accompanying drawings. In the following detailed description, numerous specific details are set forth in order to provide a thorough understanding of the various described embodiments. However, it will be apparent to one of ordinary skill in the art that the various described embodiments may be practiced without these specific details. In other instances, well-known methods, procedures, components, features, and elements have not been described in detail so as not to unnecessarily obscure aspects of the embodiments.

It is to be understood that the disclosed embodiments are merely exemplary and that various and alternative forms are possible. The figures are not necessarily to scale; some features may be exaggerated or minimized to show details of particular components. Therefore, specific structural and functional details disclosed herein are not to be interpreted as limiting, but merely as a representative basis for teaching one skilled in the art to variously employ embodiments according to the disclosure.

“One or more” and/or “at least one” includes a function being performed by one element, a function being performed by more than one element, e.g., in a distributed fashion, several functions being performed by one element, several functions being performed by several elements, or any combination of the above.

It will also be understood that, although the terms first, second, etc. are, in some instances, used herein to describe various elements, these elements should not be limited by these terms. These terms are only used to distinguish one element from another. For example, a first contact could be termed a second contact, and, similarly, a second contact could be termed a first contact, without departing from the scope of the various described embodiments. The first contact and the second contact are both contacts, but they are not the same contact.

The terminology used in the description of the various described embodiments herein is for the purpose of describing particular embodiments only and is not intended to be limiting. As used in the description of the various described embodiments and the appended claims, the singular forms “a”, “an” and “the” are intended to include the plural forms

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as well, unless the context clearly indicates otherwise. It will also be understood that the term “and/or” as used herein refers to and encompasses any and all possible combinations of one or more of the associated listed items. It will be further understood that the terms “includes,” “including,” “comprises,” and/or “comprising,” when used in this specification, specify the presence of stated features, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, steps, operations, elements, components, and/or groups thereof.

As used herein, the term “if” is, optionally, construed to mean “when” or “upon” or “in response to determining” or “in response to detecting,” depending on the context. Similarly, the phrase “if it is determined” or “if [a stated condition or event] is detected” is, optionally, construed to mean “upon determining” or “in response to determining” or “upon detecting [the stated condition or event]” or “in response to detecting [the stated condition or event],” depending on the context.

The present disclosure relates to pre-charging a low voltage stage in a direct-current-to-direct-current (DC-to-DC) voltage converter. In that regard, FIG. 1 is a simplified electrical schematic/block diagram of a non-limiting, exemplary embodiment of a DC-to-DC converter 10, which may be used in an electric vehicle (EV) or a hybrid electric vehicle (HEV).

In the embodiment shown, the converter 10 comprises a primary stage 12, a transformer 14, a secondary stage 16, and a controller 18. In that regard, it is noted that a DC-to-DC converter used in an EV or HEV may have two parallel DC-to-DC converter rails connected at an output. Only one such rail is shown in FIG. 1 for simplicity.

A high-voltage battery 20, a filter 22 (to prevent noise trespassing), a pre-charge switch 24 (shown in FIG. 1 as a metal-oxide-semiconductor field-effect transistor (MOSFET)), sensors 26, and a capacitor 28 are connected to the primary stage 12 of the DC-to-DC converter 10. The primary stage 12 comprises an H-Bridge configuration 30 of a plurality of MOSFETs 30a, 30b, 30c, 30d. The primary stage 12 may alternatively be referred to as a first or high-voltage stage or side.

A low-voltage (LV) battery 32, which may be part of a vehicle low-voltage network, sensors/filters 34, a back-to-back (B2B) switch 36 (shown in FIG. 1 as two pairs of MOSFETs provided in a back-to-back configuration), and a capacitor 38 are connected to the secondary stage of the DC-to-DC converter 10. The secondary stage 16 comprises an H-Bridge configuration 40 of a plurality of MOSFETs 40a, 40b, 40c, 40d. It is noted that the capacitor 38 may be provided for filtering of the output of the H-Bridge 40. The secondary stage 16 may alternatively be referred to as a second or low-voltage stage or side.

As is well known to those of ordinary skill, the controller 18 is configured to control and controls the ON/OFF switching operations, including duty cycle, of the transistors of the H-Bridge 30 and the H-Bridge 40 (as well as the ON/OFF operation of pre-charge switch 24 and the B2B switch 36) via switching control signals (s_p) generated by and transmitted from the controller 18 to the MOSFETs 30a, 30b, 30c, 30d of the primary stage 12 and switching control signals (s_s) generated by and transmitted from the controller 18 to the MOSFETs 40a, 40b, 40c, 40d of the secondary stage 16 according to a desired switching frequency. In such a fashion, the controller 18 operates the DC-to-DC converter 10 in a mode in order to convert the high-voltage supplied by the HV battery 20 to a low-voltage and thereby charge the LV battery 32 and/or power the vehicle low-voltage net-

work. In that regard, it is also noted that the DC-to-DC converter **10** may be bi-directional, also passing energy from LV to HV. As used herein, high-voltage may refer to a voltage greater than or equal 48 volts, while low-voltage may refer to a voltage less than or equal to 24 volts. Such a DC-to-DC converter is shown and described in detail in U.S. Pat. No. 11,502,613 B2, which is hereby incorporated herein by reference in its entirety.

Still referring to FIG. 1, at a start of the DC-to-DC converter **10**, the controller **18** closes the B2B switch **36** (via a switch control signal s_{B2B}) to enable energy flow to the low-voltage network and the LV battery **32**. However, as the capacitor **38** may be discharged at such a start of the DC-to-DC converter **10** and as the voltage at **V1** may be nearly zero (0) volts at such a start, connecting **V1** to **V2** by closing the B2B switch **36** may lead the capacitor **38** to generate a peak current. That is, in such circumstances, a peak current may flow from the LV battery **32** to the capacitor **38**. As a result, output sensors **34** detecting such a peak current may result in the controller **18** raising, generating, or determining a system overcurrent alarm and, in response, opening the B2B switch **36** (via a switch control signal s_{B2B}). Such peak current may also stress components at the secondary (i.e., low-voltage) stage **16** of the DC-to-DC converter **10**.

The present disclosure addresses such problems associated with the operation of the DC-to-DC converter **10**. In that regard, according to the present disclosure, the capacitor **38** in the secondary stage **16** is pre-charged with conversion bursts from the DC-to-DC converter **10** operated by the controller **18** in an open-loop mode to avoid undesired inrush current from the LV battery **32** at closure of the B2B switch **36** that would trigger the system over-current alarm. More specifically, before connecting the DC-to-DC converter **10** to the low-voltage network by closing the B2B-Switch **36**, the controller controls the DC-to-DC converter **10** to pre-charge the capacitor **38** connected in parallel to the H-Bridge configured transistors **40a**, **40b**, **40c**, **40d** of the secondary stage **16**. In that regard, the DC-to-DC converter **10** is activated and operated by the controller **18** in an open-loop mode, providing consecutive short pulses of energy to capacitor **38** during a selected period of time (e.g., 10 millisecond (ms)). If the voltage of the capacitor **38** fails to reach an expected level after that time period, then an anomaly at the MOSFETS **30a**, **30b**, **30c**, **30d**, **40a**, **40b**, **40c**, and/or **40d** in the H-Bridges is identified.

In that regard, FIG. 2 is a simplified electrical schematic/block diagram of a non-limiting, exemplary embodiment of a DC-to-DC converter **10**, similar to that of FIG. 1. FIG. 3 is a table of selected time periods corresponding to battery voltage ranges in a non-limiting, exemplary embodiment according to the present application. FIG. 4 is a simplified timing and signal diagram illustrating exemplary charging pulses in a non-limiting, exemplary embodiment according to the present disclosure.

With reference to FIGS. 2-4, at a start of the DC-to-DC converter **10**, the controller **18** closes the pre-charge switch **24** (via a switch control signal s_1) and keeps open the B2B switch **36** (via a switch control signal s_{B2B}) to isolate the low-voltage network and LV battery **32** from the capacitor **38**. In that regard, it is noted that the B2B switch **36** is open when the DC-to-DC converter **10** stops operation, and stays open until it is closed afterwards. The controller **18** also sets the DC-to-DC converter **10** in an open-loop mode and pre-charges the capacitor **38** to a selected, threshold, or target voltage by operating the MOSFETS **30a**, **30b**, **30c**, **30d**, of the primary stage **12** and the MOSFETS **40a**, **40b**,

40c, **40d** of the secondary stage **16**. More specifically, the controller operates the DC-to-DC controller **10** in an open-loop mode by switching control signals (s_p) generated by and transmitted from the controller **18** to the MOSFETS **30a**, **30b**, **30c**, **30d** of the primary stage **12** and switching control signals (s_s) generated by and transmitted from the controller **18** to the MOSFETS **40a**, **40b**, **40c**, **40d** of the secondary stage **16** according to a desired switching frequency (e.g., 100 kHz). In that regard, it is noted that the particular target voltage utilized, designated, or selected depends on the system involved. For example, in a system having an overvoltage alarm at 17.5 volts, the capacitor **38** may be pre-charged to a selected, threshold, or target voltage of 15.5 volts in order to avoid triggering such an overvoltage alarm.

In one embodiment of the present disclosure, such pre-charging of the capacitor **38** is controlled by the controller **18** operating the DC-to-DC converter **10** for a selected period of time, such as 10 ms, to provide controller bursts of energy of selected durations (t_1+t_2) as seen in FIG. 4. In such an embodiment, t_1 is initially set to 500 nanoseconds (ns) and t_2 is set to a constant 500 ns. At a selected time such as every 1 ms, the controller **18** increases t_1 in value from an initial duration of 500 ns to a maximum duration of 4 microseconds (μ s), while $V1$ (i.e., the continuous DC voltage level or charge of the capacitor **38**) increases up to the selected, threshold, or target voltage. In that regard, it is noted that alternative times t_1 and t_2 may be utilized, designated, or selected depending on the system involved.

In that regard, the rate of such an increase depends on the voltage of the HV battery **20**. As seen in FIG. 3, according to one embodiment of the present disclosure, for an HV battery **20** having a voltage in the range of 430 to 470 volts DC, the rate of increase of t_1 by the controller **18** is 50 ns/ms. For an HV battery **20** having a voltage in the range of 290 to 430 volts DC, the rate of increase of t_1 by the controller **18** is 470 ns/ms to 50 ns/ms. For an HV battery **20** having a voltage in the range of 250 to 300 volts DC, the rate of increase of t_1 by the controller **18** is 100 ns/ms. Thus, over a 10 ms time frame, the duration t_1 of an initial energy burst produced by the controller **18** controlling the DC-to-DC converter **10** starts at 500 ns, and then every 1 ms the duration t_1 of a subsequent energy burst increases in value up to a maximum of 4 μ s at a rate shown in FIG. 3 depending on the voltage of the HV battery **20**. Once again, it is noted that alternative voltages, voltage ranges, times and/or rates may be utilized, designated, or selected depending on the system involved.

If the voltage at **V1** (i.e., the continuous DC voltage level or charge of the capacitor **38**) has not reached the selected, threshold, or target voltage after 10 ms, then the controller **18** determines that one or more of the MOSFETS **40a**, **40b**, **40c**, **40d** of the secondary stage **16** have been damaged. In such a fashion, the present disclosure provides a MOSFET health monitoring mechanism for the DC-to-DC converter **10**. Alternatively, when the voltage at **V1** (i.e., the continuous DC voltage level or charge of the capacitor **38**) reaches the selected, threshold, or target voltage, then the controller **18** closes the B2B switch **36** (via a switch control signal s_{B2B}) to connect the capacitor **38** to the low-voltage network and the LV battery **32**. In such a fashion, the controller **18** thereby ends the pre-charge operation and begins normal mode operation of the DC-to-DC converter **10**.

Exemplary timing for switch operation and the production of primary and secondary voltages in a DC-to-DC converter is described in detail in U.S. Patent Application Publication No. 2022/077789 A1, which is hereby incorporated herein by reference in its entirety. An example of the relationship

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between the switched voltage V_p of the primary stage **12**, the switched voltage V_s of the secondary stage **16**, and the current I_s of the secondary stage **16**, as controlled by the controller **18** according to the present disclosure, is shown in the timing and signal diagram of FIG. 4.

As seen therein, according to one embodiment of the present disclosure, the controller **18** operates the MOSFETS **30a**, **30b**, **30c**, **30d** of the primary stage **12** and the MOSFETS **40a**, **40b**, **40c**, **40d** of the secondary stage **16** to produce bursts of energy, in particular a primary switched voltage V_p for a duration or time t_1 as previously described and a secondary switched voltage V_s for a duration or time (t_1+t_2) as also previously described. The time between switching on the voltage at the primary stage **12** and then switching on the secondary stage **16** by the controller **18** is shown therein as phase-shift time, ps . In one embodiment according to the present disclosure, the phase-shift time is -170 ns. Once again, however, it is noted that an alternative phase-shift time may be utilized, designated, or selected depending on the system involved.

As seen in FIG. 4, according to the present disclosure, the controller **18** controls the MOSFETS **30a**, **30b**, **30c**, **30d** of the primary stage **12** and the MOSFETS **40a**, **40b**, **40c**, **40d** of the secondary stage **16** to add a gap or dead time between switched energy pulses or bursts, in contrast to switching during normal mode operation where no dead time occurs between switched energy pulses or bursts. In that regard, the period of the energy pulses or bursts (i.e., the time from the start of one pulse to the start of the next pulse) is fixed based on the switching frequency (e.g., 100 kHz) employed by the controller **18** for operation of the MOSFETS **30a**, **30b**, **30c**, **30d** of the primary stage **12** and the MOSFETS **40a**, **40b**, **40c**, **40d** of the secondary stage **16**. The dead time between energy pulses or bursts thus decreases as the duty cycle of (t_1+t_2) increases according to the present disclosure as described herein.

As those skilled in the art will understand, the controller **18**, as well as any other component, system, subsystem, unit, module, circuit, stage, interface, sensor, device, or the like described herein may individually, collectively, or in any combination comprise appropriate circuitry, such as one or more appropriately programmed processors (e.g., one or more microprocessors including central processing units (CPU)) and associated memory, which may include stored operating system software, firmware, and/or application software executable by the processor(s) for controlling operation thereof, any component, system, subsystem, unit, module, circuit, stage, interface, sensor, device, or the like described herein, and/or for performing the particular algorithm or algorithms represented by the various methods, functions and/or operations described herein, including interaction between and/or cooperation with each other.

Item 1: According to an embodiment, the present disclosure provides a converter comprising a primary stage comprising a plurality of primary stage switches, a secondary stage comprising a plurality of secondary stage switches and a capacitor, and a controller that isolates the secondary stage from a second battery, and that controls switching of the plurality of primary stage switches and the plurality of secondary stage switches to provide a plurality of bursts of energy from a first battery to the capacitor to pre-charge the capacitor to a target voltage, wherein after an initial burst of the plurality of bursts having an initial duration, a subsequent burst of the plurality of bursts provided has a duration equal to a duration of a preceding burst plus a time period based on a voltage of the first battery.

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Item 2: In another embodiment, the present disclosure provides the converter according to Item 1 wherein the controller controls the plurality of primary stage switches and the plurality of secondary stage switches to pre-charge the capacitor for a selected period of time.

Item 3: In another embodiment, the present disclosure provides the converter according to Item 1 or Item 2 wherein the controller controls the plurality of primary stage switches and plurality of secondary stage switches to provide the plurality of bursts of energy until the capacitor reaches a charge equal to the target voltage.

Item 4: In another embodiment, the present disclosure provides the converter according to any of Items 1-3 wherein the controller controls the plurality of primary stage switches and the plurality of secondary stage switches to produce the subsequent burst of the plurality of bursts at a selected time after the initial burst of the plurality of bursts.

Item 5: In another embodiment, the present disclosure provides the converter according to any of Items 1-4 wherein the controller controls the plurality of primary stage switches and the plurality of secondary stage switches to produce another subsequent burst of the plurality of bursts at the selected time after the subsequent burst of the plurality of bursts.

Item 6: In another embodiment, the present disclosure provides the converter according to any of Items 1-5 wherein a duration of another subsequent burst is limited to a maximum duration when a sum of the time period dependent upon the voltage of the first battery and a duration of a burst immediately preceding the another subsequent burst exceeds the maximum duration.

Item 7: In another embodiment, the present disclosure provides the converter according to any of Items 1-6 wherein the time period based on the voltage of the first battery has a first value when the voltage of the first battery falls within a first voltage range, and wherein the time period based on the voltage of the first battery has a second value when the voltage of the first battery falls within a second voltage range different than the first voltage range.

Item 8: In another embodiment, the present disclosure provides the converter according to any of Items 1-7 wherein the controller connects the secondary stage to the second battery in response to a charge of the capacitor reaching the target voltage.

Item 9: In another embodiment, the present disclosure provides the converter according to any of Items 1-8 wherein a failure of the capacitor to reach a charge equal to the target voltage within a threshold period of time indicates damage to at least one of the plurality of secondary stage switches.

Item 10: In another embodiment, the present disclosure provides the converter according to any of Items 1-9 wherein the first battery has a voltage of greater than or equal to 48 volts and the second battery has a voltage of less than or equal to 24 volts.

Item 11: According to an embodiment, the present disclosure provides a non-transitory computer readable medium having stored computer executable instructions for controlling a converter comprising (i) a primary stage comprising a plurality of primary stage switches, (ii) a secondary stage comprising a plurality of secondary stage switches and a capacitor, and (iii) a controller, wherein execution of the instructions causes the controller to isolate the secondary stage from a second battery, and control switching of the plurality of primary stage switches and the plurality of secondary stage switches to provide a plurality of bursts of energy from a first battery to the capacitor to pre-charge the

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capacitor to a target voltage, wherein after an initial burst of the plurality of bursts having an initial duration, a subsequent burst of the plurality of bursts provided has a duration equal to a duration of a preceding burst plus a time period based on a voltage of first battery.

Item 12: In another embodiment, the present disclosure provides the non-transitory computer readable medium according to Item 11 wherein execution of the instructions further causes the controller to control the plurality of primary stage switches and the plurality of secondary stage switches to pre-charge the capacitor for a selected period of time.

Item 13: In another embodiment, the present disclosure provides the non-transitory computer readable medium according to Item 11 or Item 12 wherein execution of the instructions further causes the controller to control the plurality of primary stage switches and plurality of secondary stage switches to provide the plurality of bursts of energy until the capacitor reaches a charge equal to the target voltage.

Item 14: In another embodiment, the present disclosure provides the non-transitory computer readable medium according to any of Items 11-13 wherein execution of the instructions further causes the controller to control the plurality of primary stage switches and the plurality of secondary stage switches to produce the subsequent burst of the plurality of bursts at a selected time after the initial burst of the plurality of bursts.

Item 15: In another embodiment, the present disclosure provides the non-transitory computer readable medium according to any of Items 11-14 wherein execution of the instructions further causes the controller to control the plurality of primary stage switches and the plurality of secondary stage switches to produce another subsequent burst of the plurality of bursts at the selected time after the subsequent burst of the plurality of bursts.

Item 16: In another embodiment, the present disclosure provides the non-transitory computer readable medium according to any of Items 11-15 wherein a duration of another subsequent burst is limited to a maximum duration when a sum of the time period dependent upon the voltage of the first battery and a duration of a burst immediately preceding the another subsequent burst exceeds the maximum duration.

Item 17: In another embodiment, the present disclosure provides the non-transitory computer readable medium according to any of Items 11-16 wherein the time period based on the voltage of the first battery has a first value when the voltage of the first battery falls within a first voltage range, and wherein the time period based on the voltage of the first battery has a second value when the voltage of the first battery falls within a second voltage range different than the first voltage range.

Item 18: In another embodiment, the present disclosure provides the non-transitory computer readable medium according to any of Items 11-17 wherein execution of the instructions further causes the controller to connect the secondary stage to the second battery in response to a charge of the capacitor reaching the target voltage.

Item 19: In another embodiment, the present disclosure provides the non-transitory computer readable medium according to any of Items 11-18 wherein a failure of the capacitor to reach a charge equal to the target voltage within a threshold period of time indicates damage to at least one of the plurality of secondary stage switches.

Item 20: In another embodiment, the present disclosure provides the non-transitory computer readable medium

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according to any of Items 11-19 wherein the first battery has a voltage of greater than or equal to 48 volts and the second battery has a voltage of less than or equal to 24 volts.

The present disclosure thus provides for a DC-to-DC converter wherein a low-voltage output overcurrent alarm may be provided without undesired triggering. The present disclosure also provides a DC-to-DC converter wherein component stress is reduced, thereby increasing component and converter lifespans. The present disclosure further provides a DC-to-DC converter wherein MOSFET malfunction is monitored before full power transfer is initiated. Still further, the present disclosure may be implemented without hardware additions to a DC-to-DC converter, utilizing a software concept for control of a DC-to-DC converter by a programmed controller.

While exemplary embodiments are described above, it is not intended that these embodiments describe all possible forms according to the disclosure. In that regard, the words used in the specification are words of description rather than limitation, and it is understood that various changes may be made without departing from the spirit and scope of the disclosure. Additionally, unless the context clearly indicates otherwise, the various features, elements, components, methods, procedures, steps, and/or functions of various implementing embodiments may be combined or utilized in any combination or combinations and/or may be performed in any order other than those specifically described herein to form further embodiments according to the present disclosure.

What is claimed is:

1. A converter comprising:

a primary stage comprising a plurality of primary stage switches;

a secondary stage comprising a plurality of secondary stage switches and a capacitor; and

a controller that isolates the secondary stage from a second battery, and that controls switching of the plurality of primary stage switches and the plurality of secondary stage switches to provide a plurality of bursts of energy from a first battery to the capacitor to pre-charge the capacitor to a target voltage;

wherein after an initial burst of the plurality of bursts having an initial duration, a subsequent burst of the plurality of bursts provided has a duration equal to a duration of a preceding burst plus a time period based on a voltage of the first battery.

2. The converter according to claim 1 wherein the controller controls the plurality of primary stage switches and the plurality of secondary stage switches to pre-charge the capacitor for a selected period of time.

3. The converter according to claim 1 wherein the controller controls the plurality of primary stage switches and plurality of secondary stage switches to provide the plurality of bursts of energy until the capacitor reaches a charge equal to the target voltage.

4. The converter according to claim 1 wherein the controller controls the plurality of primary stage switches and the plurality of secondary stage switches to produce the subsequent burst of the plurality of bursts at a selected time after the initial burst of the plurality of bursts.

5. The converter according to claim 4 wherein the controller controls the plurality of primary stage switches and the plurality of secondary stage switches to produce another subsequent burst of the plurality of bursts at the selected time after the subsequent burst of the plurality of bursts.

6. The converter according to claim 1 wherein a duration of another subsequent burst is limited to a maximum dura-

tion when a sum of the time period dependent upon the voltage of the first battery and a duration of a burst immediately preceding the another subsequent burst exceeds the maximum duration.

7. The converter according to claim 1 wherein the time period based on the voltage of the first battery has a first value when the voltage of the first battery falls within a first voltage range, and wherein the time period based on the voltage of the first battery has a second value when the voltage of the first battery falls within a second voltage range different than the first voltage range.

8. The converter according to claim 1 wherein the controller connects the secondary stage to the second battery in response to a charge of the capacitor reaching the target voltage.

9. The converter according to claim 1 wherein a failure of the capacitor to reach a charge equal to the target voltage within a threshold period of time indicates damage to at least one of the plurality of secondary stage switches.

10. The converter according to claim 1 wherein the first battery has a voltage of greater than or equal to 48 volts and the second battery has a voltage of less than or equal to 24 volts.

11. A non-transitory computer readable medium having stored computer executable instructions for controlling a converter comprising (i) a primary stage comprising a plurality of primary stage switches, (ii) a secondary stage comprising a plurality of secondary stage switches and a capacitor, and (iii) a controller, wherein execution of the instructions causes the controller to:

isolate the secondary stage from a second battery; and control switching of the plurality of primary stage switches and the plurality of secondary stage switches to provide a plurality of bursts of energy from a first battery to the capacitor to pre-charge the capacitor to a target voltage;

wherein after an initial burst of the plurality of bursts having an initial duration, a subsequent burst of the plurality of bursts provided has a duration equal to a duration of a preceding burst plus a time period based on a voltage of first battery.

12. The non-transitory computer readable medium according to claim 11 wherein execution of the instructions further causes the controller to control the plurality of primary stage switches and the plurality of secondary stage switches to pre-charge the capacitor for a selected period of time.

13. The non-transitory computer readable medium according to claim 11 wherein execution of the instructions

further causes the controller to control the plurality of primary stage switches and plurality of secondary stage switches to provide the plurality of bursts of energy until the capacitor reaches a charge equal to the target voltage.

14. The non-transitory computer readable medium according to claim 11 wherein execution of the instructions further causes the controller to control the plurality of primary stage switches and the plurality of secondary stage switches to produce the subsequent burst of the plurality of bursts at a selected time after the initial burst of the plurality of bursts.

15. The non-transitory computer readable medium according to claim 14 wherein execution of the instructions further causes the controller to control the plurality of primary stage switches and the plurality of secondary stage switches to produce another subsequent burst of the plurality of bursts at the selected time after the subsequent burst of the plurality of bursts.

16. The non-transitory computer readable medium according to claim 11 wherein a duration of another subsequent burst is limited to a maximum duration when a sum of the time period dependent upon the voltage of the first battery and a duration of a burst immediately preceding the another subsequent burst exceeds the maximum duration.

17. The non-transitory computer readable medium according to claim 11 wherein the time period based on the voltage of the first battery has a first value when the voltage of the first battery falls within a first voltage range, and wherein the time period based on the voltage of the first battery has a second value when the voltage of the first battery falls within a second voltage range different than the first voltage range.

18. The non-transitory computer readable medium according to claim 11 wherein execution of the instructions further causes the controller to connect the secondary stage to the second battery in response to a charge of the capacitor reaching the target voltage.

19. The non-transitory computer readable medium according to claim 11 wherein a failure of the capacitor to reach a charge equal to the target voltage within a threshold period of time indicates damage to at least one of the plurality of secondary stage switches.

20. The non-transitory computer readable medium according to claim 11 wherein the first battery has a voltage of greater than or equal to 48 volts and the second battery has a voltage of less than or equal to 24 volts.

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